Name: Kanishka Sharma CU24250095 Sec:B Roll No: 26

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder, StandardScaler

print("liberaries are installed ") liberaries are installed

df =
pd.read_csv("/workspaces/Mastery_in_DataAnalytics_and_Visualizations_a
nd_Career_Advancemen/train-selected-columns.csv")

print("First 5 rows:")
print(df.head())

print("\nDataset Information:")
df.info()

print("\nStatistical Description:")
print(df.describe())
```

```
First 5 rows:
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age
SibSp  \
                                                                  0
                             Braund, Mr. Owen Harris    male  22.0 1
                                                                  1
 Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0 1
                                                                  2
                              Heikkinen, Miss. Laina  female  26.0 0
                                                                  3
        Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0 1
                                                                  4
                            Allen, Mr. William Henry    male  35.0 0

   Parch            Ticket     Fare
0      0         A/5 21171   7.2500
1      0          PC 17599  71.2833

2      0  STON/O2. 3101282   7.9250
```

```
3      0            113803  53.1000
4      0            373450   8.0500

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    str
4   Sex          891 non-null    str
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    str     9   Fare         891 non-
    null    float64 dtypes: float64(2), int64(5), str(3) memory
    usage: 69.7 KB

Statistical Description:
       PassengerId    Survived      Pclass         Age       SibSp  \
count    891.000000  891.000000  891.000000  714.000000  891.000000
mean     446.000000    0.383838    2.308642   29.699118    0.523008
std      257.353842    0.486592    0.836071   14.526497    1.102743
min        1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208  std
0.806057   49.693429  min
0.000000    0.000000  25%
0.000000    7.910400
50%      0.000000   14.454200  75%
0.000000   31.000000  max
6.000000  512.329200

print("Dataset Information:")
df.info()

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
0     PassengerId  891 non-null    int64
1     Survived     891 non-null    int64
2     Pclass       891 non-null    int64
3     Name         891 non-null    str
4     Sex          891 non-null    str
5     Age          714 non-null    float64
6     SibSp        891 non-null    int64
7     Parch        891 non-null    int64
8     Ticket       891 non-null    str     9   Fare         891 non-
null    float64 dtypes: float64(2), int64(5), str(3) memory usage:
69.7 KB

print("Statistical Description:")
df.describe()
Statistical
Description:
       PassengerId
Survived          Pclass
Age       SibSp  \
count          891.000000
891.000000    891.000000
714.000000    891.000000
mean           446.000000
0.383838        2.308642
29.699118       0.523008
std            257.353842
0.486592        0.836071
14.526497       1.102743
min              1.000000
0.000000        1.000000
0.420000    0.000000 25%
223.500000      0.000000
2.000000       20.125000
0.000000
50%            446.000000
0.000000        3.000000
28.000000       0.000000
75%            668.500000
1.000000        3.000000
38.000000       1.000000
max            891.000000
1.000000        3.000000
80.000000    8.000000
```

```
Parch       Fare
count
891.000000
891.000000  mean
0.381594
32.204208    std
0.806057
49.693429    min
0.000000
0.000000     25%
0.000000
7.910400
50%     0.000000
14.454200    75%
0.000000
31.000000    max
6.000000
512.329200
```

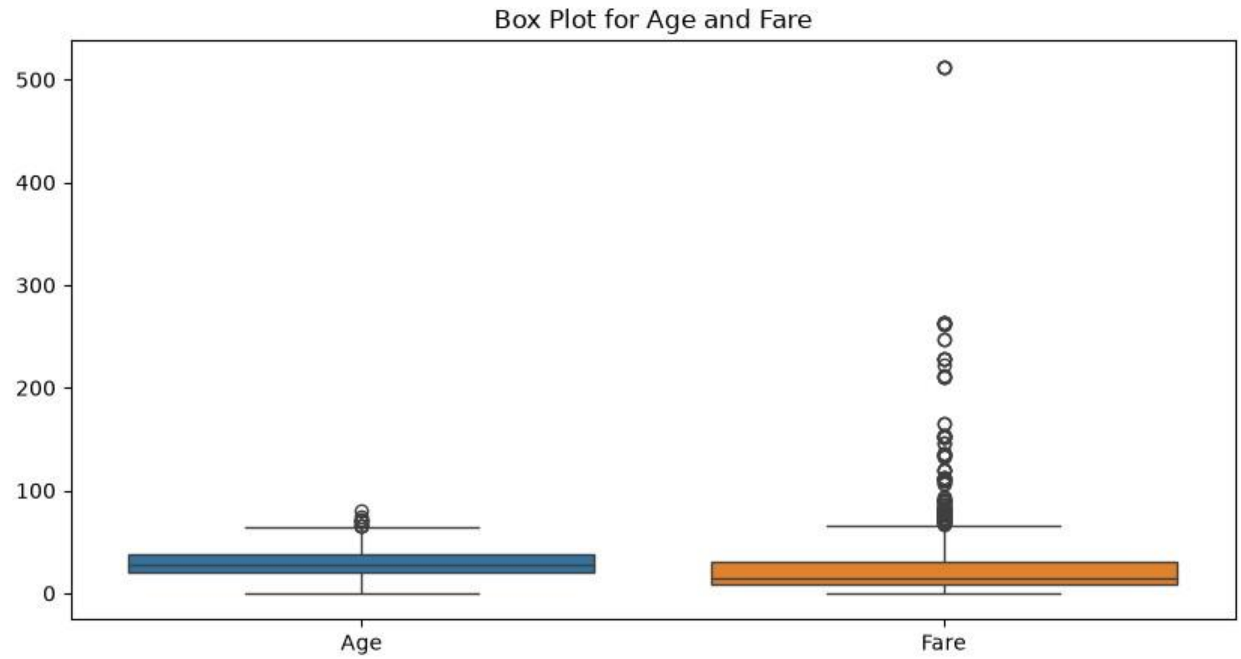
```python
print("Dataset Shape:")
print(df.shape)
```

```
Dataset Shape:
(891, 10)
```

```python
print("Missing Values:")
print(df.isnull().sum())
```

```
Missing Values:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
dtype: int64
print("Missing Value Percentage:")
print((df.isnull().sum() / len(df)) * 100)
Missing Value Percentage:
PassengerId     0.00000
Survived        0.00000
Pclass          0.00000
Name            0.00000
Sex             0.00000
Age            19.86532
SibSp           0.00000
Parch           0.00000
Ticket          0.00000
Fare            0.00000
dtype: float64
print("Duplicate records before removal:")
print(df.duplicated().sum())
df.drop_duplicates(inplace=True)

print("Duplicate records after removal:")
print(df.duplicated().sum())

Duplicate records before removal:
0 Duplicate records after
removal:
0

df =
pd.read_csv("/workspaces/Mastery_in_DataAnalytics_and_Visualizations_a
nd_Career_Advancemen/train-selected-columns.csv")
print(df.columns.tolist())
['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
'Parch', 'Ticket', 'Fare']
```

```
df.to_csv("cleaned_train.csv", index=False)
print("Cleaned dataset saved successfully!")
Cleaned dataset saved successfully!
# Check duplicate records
print("Duplicate records before removal:")
print(df.duplicated().sum())

# Remove duplicate records
df.drop_duplicates(inplace=True)

# Check duplicates again
print("\nDuplicate records after removal:")
print(df.duplicated().sum())

Duplicate records before removal:
0

Duplicate records after removal:
0 # Step 7: Identify outliers using Box Plot and

IQR

import matplotlib.pyplot as plt
import seaborn as sns

# Box plots
plt.figure(figsize=(10, 5))
sns.boxplot(data=df[['Age', 'Fare']])
plt.title("Box Plot for Age and Fare")
plt.show()

# IQR for Age
Q1 = df['Age'].quantile(0.25)
Q3 = df['Age'].quantile(0.75)

IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

# Remove Age outliers
df = df[(df['Age'].isna()) |
        ((df['Age'] >= lower_limit) & (df['Age'] <= upper_limit))]

print("Outliers in Age treated successfully.")
print("New dataset shape:", df.shape)
```


Box Plot for Age and Fare
500
400
300
200
100
0
Age
Fare

```
Outliers in Age treated successfully.
New dataset shape: (880, 10)

# Step 8: Normalization

from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
 Age and Fare normalized successfully.
c         Age      Fare                  form(df[['Age', 'Fare']])

print("Age and Fare normalized successfully.")
print(df[['Age', 'Fare']].head())

 4  0.543882  0.015713                  # Step 9: Feature Engineering

                                        0  0.339415  0.014151  # Create
                                        1  0.591066  0.139136  Family
Size                                    2  0.402328  0.015469
                                        3  0.543882  0.103644
                      df['FamilySize'] = df['SibSp'] + df['Parch'] + 1

# Create Age Group
df['AgeGroup'] = pd.cut(
df['Age'],
```

```
    bins=[0, 18, 35, 60, 100],
    labels=['Child', 'Young', 'Adult', 'Senior']
)

print("New features created successfully.")
print(df[['FamilySize', 'AgeGroup']].head())
New features created successfully.
   FamilySize AgeGroup
0           2    Child
1           2    Child
2           1    Child
3           2    Child
4           1    Child
# Step 10: Save the cleaned dataset

df.to_csv('titanic_cleaned.csv', index=False)

print("Cleaned Titanic dataset saved successfully!")

Cleaned Titanic dataset saved successfully!
```